



Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ APT: Arbitrage Pricing Theory
- ▣ BAB: Betting Against Beta
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CML: Capital Market Line
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMM: Generalized Method of Moments
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HJ: Hansen–Jagannathan (distance)
- ▣ HML: High Minus Low (value factor)
- ▣ IPCA: Instrumented Principal Component Analysis
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ MOM: MOMentum factor
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares

Acronyms used in this material (2/2)

- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ PC1: first Principal Component
- ▣ PC2: second Principal Component
- ▣ PC3: third Principal Component
- ▣ PCA: Principal Component Analysis
- ▣ RF: Risk-Free rate
- ▣ RMSE: Root Mean Squared Error
- ▣ RMW: Robust Minus Weak (profitability factor)
- ▣ ROE: Return On Equity
- ▣ RP-PCA: Risk-Premium Principal Component Analysis
- ▣ SDF: Stochastic Discount Factor
- ▣ SMB: Small Minus Big (size factor)
- ▣ SML: Security Market Line
- ▣ UMD: Up Minus Down (momentum factor)
- ▣ US: United States

Outline

Motivation

From Markowitz to the CAPM

Estimating Beta

Testing the CAPM

Fama–MacBeth Regressions

Multifactor Models

The Factor Zoo and Multiple Testing

Latent Factors: PCA and IPCA

Factor Investing

Case Study: Is There a Replication Crisis in Finance?

AI for Scientific Discovery

Conclusions

References

What Is a Factor Model?

Definition

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \dots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}f_{1,t} + \dots + \beta_{i,K}f_{K,t} + \varepsilon_{i,t}$$

- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k ; $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock

Key message of this chapter

- Expected returns should be explained by **exposures to priced risks**, not by the name of the asset; testing this claim is the core of empirical asset pricing

Why Factor Models Matter

- ▣ **Cost of capital:** the return investors require from a firm or project
- ▣ **Performance evaluation:** did a fund beat the market, or did it only carry more risk?
- ▣ **Risk management:** N assets need $N(N + 1)/2$ covariances; K factors reduce this drastically
- ▣ **Portfolio construction:** factor exposures can be bought directly (factor ETFs, Section 8)
- ▣ **Asset pricing:** which risks are rewarded, and why? (textbook treatment: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 5–6)

Data Used in This Chapter

Series	Content	Period
Fama–French factors	market, size, value, profitability, investment	1963–2026
Momentum factor	winners minus losers	1963–2026
25 size $\times$ B/M portfolios	US stocks sorted on two characteristics	1963–2026
US sector ETFs	11 sectors of the S&P 500	1999–2026
Factor ETFs	momentum, value, quality, min. volatility, size	2013–2026
BVB blue chips, BET	11 Romanian stocks and their index	2015–2026

- Factor and portfolio returns: [Kenneth French Data Library](#)
- Prices: repository folder data/market, up to 18 September 2026

The Data

- ▣ **ETFs** used in the chapter
 - ▶ Sector ETFs: XLK (technology), XLF (financials), XLE (energy), XLU (utilities), ...; SPY tracks the S&P 500
- ▣ **B/M**: book-to-market ratio, accounting value of equity divided by its market value
- ▣ **BET**: the benchmark index of the BVB
- ▣ Blue chips used: TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL (full names on the chart)
- ▣ ETFs and stocks: dividend- and split-adjusted prices; indices: closing values (Chapter 1)

Excess Returns and the Risk-Free Rate

- **Risk-free rate** $R_{f,t}$: return on an asset with no default risk over the period
 - ▶ US: one-month Treasury bill rate (column RF of the Fama–French files); mean 4.35% a year since 1963
- **Excess return**: $R_{i,t}^e = R_{i,t} - R_{f,t}$, the reward for bearing risk
- **Market excess return** (Mkt-RF): value-weighted US market minus $R_{f,t}$
- Assumption for the BVB: daily $R_{f,t}$ is treated as constant
 - ▶ A constant cancels in covariances, so betas are unchanged; only the intercept shifts
- Regressions on Fama–French factors use simple returns (the factors are simple returns); rolling, BVB and PCA analyses use log returns (Chapter 1)

Prerequisites Recap: Mean–Variance and the Tangency Portfolio

- **Markowitz (1952)**: with weights w , expected excess returns μ and covariance Σ , the frontier solves

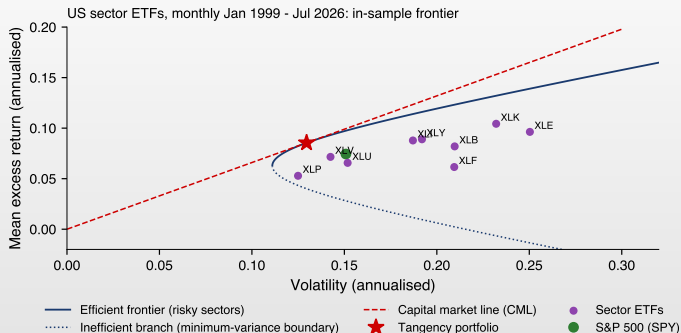
$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = \mu^*, \quad w^\top \mathbf{1} = 1$$

- **Sharpe ratio** $SR_p = \mathbb{E}[R_p - R_f] / \sigma_p$; with a risk-free asset the best risky portfolio is the **tangency portfolio**

$$w_{\text{tan}} = \Sigma^{-1} \mu / \mathbf{1}^\top \Sigma^{-1} \mu$$

- **CML**: mixes of R_f and the tangency portfolio, $\mu_p = SR_{\text{tan}} \sigma_p$; negative weights are **short positions**
- From the tangency portfolio to the beta pricing equation: Seminar A1; estimation error and shrinkage: Chapter 4

The Efficient Frontier of US Sectors



□ Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ($T = 331$ months)

Reading the Frontier

- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49
 - ▶ Here the market is **not** on the frontier, as the CAPM would require
- Tangency weights: -0.37 in financials, -0.21 in materials, $+0.37$ in health care
- Caveat: the frontier is **in-sample**; estimated means are very noisy
 - ▶ Out-of-sample optimisation and covariance shrinkage: Chapter 4

The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in Economics
1990

Photo: Larry D. Moore (2007); CC BY 4.0;
[Wikimedia Commons](#)

- **CAPM**: Capital Asset Pricing Model, [Sharpe \(1964\)](#) and [Lintner \(1965\)](#)
- Question: if every investor holds a mean–variance efficient portfolio, what are equilibrium prices?
- Answer: everyone holds the same tangency portfolio, so it must be the **market portfolio**
- [Black \(1972\)](#): a version without a risk-free asset (the zero-beta CAPM)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

The CAPM Equation

- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_m] - R_f), \quad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

- $\mathbb{E}[R_m] - R_f$: **market risk premium**; β_i : sensitivity of asset i to the market
- Main assumptions
 - Mean–variance investors with identical beliefs, one period
 - Unlimited borrowing and lending at R_f ; no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns
- SML** (Security Market Line): every asset lies on the line (beta, mean) with slope equal to the premium; the CML contains only efficient portfolios. The SML is the testable statement (Section 4)

Recap: From Markowitz to the CAPM

- Mean–variance investors hold the tangency portfolio combined with the risk-free asset
- In equilibrium this portfolio is the market, and expected excess returns are β_i times the premium
- CML: efficient portfolios in (volatility, mean); SML: every asset in (beta, mean)
- Only systematic risk is rewarded under the CAPM
- In our sector data the market proxy is far from the in-sample frontier
- CAPM cost of capital needs an estimated beta and an estimated premium: the error of the premium dominates

Prerequisites Recap: the Market Model and HAC Errors

- ▣ Market model $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$; **OLS** (Ordinary Least Squares): $\hat{\beta}_i = \widehat{\text{Cov}}(R_i^e, R_m^e) / \widehat{\text{Var}}(R_m^e)$
- ▣ Variance split $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2$: systematic (priced under the CAPM) plus idiosyncratic (diversifiable); R^2 is the systematic share
- ▣ Volatility clustering (Chapter 1) makes residuals heteroskedastic: use **HAC** errors, [Newey & West \(1987\)](#), with Bartlett weights
- ▣ Lag rule $L = \lfloor 4(T/100)^{2/9} \rfloor$: $T = 331$ months gives $L = 5$; $T = 757$ gives $L = 6$
- ▣ Same point estimates as OLS, only the standard errors change

Worked Example: Beta of the Technology ETF

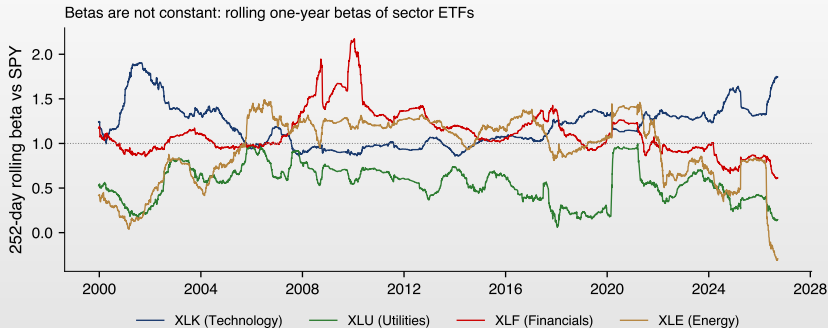
	Estimate	Classical s.e.	Newey–West s.e.
α (annualised)	0.33%	2.27%	2.45%
β	1.277	0.041	0.082

- ▣ XLK on the US market factor Mkt-RF, monthly excess returns 1999–2026, $T = 331$, $R^2 = 0.742$
- ▣ Newey–West with $L = \lfloor 4(3.31)^{2/9} \rfloor = \lfloor 5.2 \rfloor = 5$ lags
- ▣ The robust standard error of beta is **twice** the classical one
- ▣ Alpha is not statistically distinguishable from zero: $t = 0.33/2.45 = 0.13$

Estimation Choices in Practice

- ▣ **Frequency:** daily data give more observations but suffer from asynchronous trading
- ▣ **Window:** 3–5 years of monthly data or 1–2 years of daily data are common
- ▣ **Market proxy:** S&P 500 (SPY) for US assets, BET for Romanian stocks
- ▣ **Data cleaning:** unadjusted capital events create fake returns
 - ▶ Example: a daily log return of -6.6 for DIGI in May 2017; such days are removed
- ▣ Always report the sample, the frequency and the standard error of beta

Betas Are Not Constant



- Rolling beta: OLS beta on the last 252 daily log returns, recomputed every day

Reading the Rolling Betas

- ▣ Financials (XLF): from 0.60 to 2.17; the peak follows the 2008–2009 banking crisis
- ▣ Technology (XLK): from 0.86 to 1.90; high in the dot-com years, 1.75 at the end of the sample
- ▣ Energy (XLE): from -0.31 to 1.49; negative at the end, when oil moved against the market
- ▣ Utilities (XLU): from 0.06 to 1.00; the classic defensive sector
- ▣ A single full-sample beta hides these regimes; conditional models: Chapters 5–6

Shrinking Betas: Blume and Vasicek

- ▣ **Regression to the mean:** extreme estimated betas are partly noise and move towards 1 later
- ▣ **Blume (1971):** regress later betas on earlier betas; the practitioner rule is

$$\beta^{\text{Blume}} = 0.33 + 0.67 \hat{\beta}$$

- ▣ **Vasicek (1973):** Bayesian average of the estimate and the cross-sectional prior $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i) \bar{\beta}, \quad w_i = \frac{\sigma_{\beta}^2}{\sigma_{\beta}^2 + \text{se}(\hat{\beta}_i)^2}$$

- ▣ β_i^V is the posterior mean under a Normal prior $\beta_i \sim N(\bar{\beta}, \sigma_{\beta}^2)$ and $\hat{\beta}_i | \beta_i \sim N(\beta_i, \text{se}_i^2)$ (Seminar A3)
- ▣ σ_{β}^2 is the variance of the **true** betas: $\text{Var}_{cs}(\hat{\beta}) = \sigma_{\beta}^2 + \overline{\text{se}^2}$, so the empirical-Bayes prior is $\sigma_{\beta}^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$

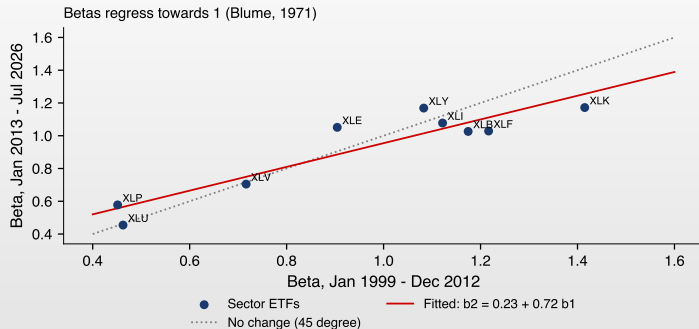
Worked Example: Vasicek Beta of XLK

- ▣ Sample January 1999 – December 2012, monthly, 9 sector ETFs
- ▣ XLK: $\hat{\beta} = 1.415$, $\text{se}(\hat{\beta}) = 0.139$ (Newey–West)
- ▣ Prior mean $\bar{\beta} = 0.949$; prior variance $\sigma_{\beta}^2 = 0.116 - 0.008 = 0.108$
 - ▶ 0.116: variance of the 9 estimated betas; 0.008: their average squared standard error
- ▣ Weight: $w = 0.108 / (0.108 + 0.139^2) = 0.849$

$$\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$$

- ▣ The realised beta in 2013–2026 was 1.172: shrinkage moved the estimate in the right direction

Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)

Reading the Beta Regression

- ▣ Fitted line: $\hat{\beta}_{\text{later}} = 0.23 + 0.72 \hat{\beta}_{\text{earlier}}$, very close to Blume's $0.33 + 0.67 \hat{\beta}$
- ▣ **RMSE** (Root Mean Squared Error) of the forecast of later betas

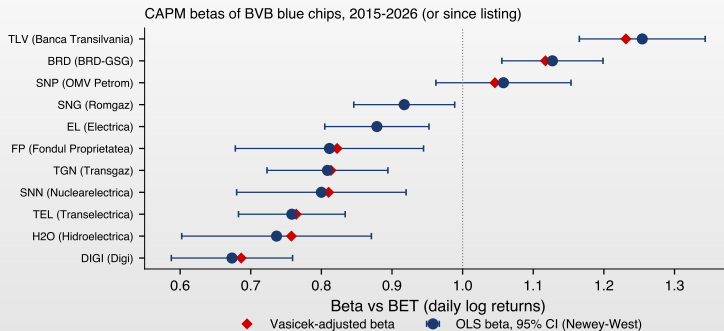
$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i (\text{forecast}_i - \hat{\beta}_{i,\text{later}})^2}$$

- ▣ Raw beta: 0.135; Blume rule: 0.109; Vasicek: 0.114
- ▣ Shrinkage cuts the forecast error by 15–20%
- ▣ Only 9 sectors: the lesson is qualitative

Romania: Betas of the BVB Blue Chips

- ▣ Regression of daily log returns of each stock on daily log returns of BET
- ▣ Sample from 6 January 2015 or from the listing date (DIGI 2017, H2O 2023)
- ▣ Risk-free rate: assumed constant at daily frequency, so it does not affect beta
- ▣ Days with $|r_t| > 0.5$ (capital events not adjusted in prices) are removed: one day for DIGI, two for FP
- ▣ Newey–West 95% confidence intervals and Vasicek-adjusted betas (empirical-Bayes prior from the 11 stocks)

CAPM Betas on the Bucharest Stock Exchange



□ Blue dots: OLS betas with 95% intervals; red diamonds: Vasicek-adjusted betas

Reading the BVB Betas

- ▣ Banks are the most market-sensitive: Banca Transilvania 1.25, BRD 1.13
- ▣ OMV Petrom 1.06; Romgaz 0.92; utilities and telecom below: Transelectrica 0.76, Digi 0.67
- ▣ R^2 : 0.59 for TLV and 0.52 for BRD, but only 0.21 for DIGI
- ▣ Large index weights (TLV, SNP, BRD) mechanically raise their R^2 against BET
- ▣ Vasicek changes little here: the daily estimates are precise (se $\approx$ 0.04–0.07)

Recap: Estimating Beta

- ▣ Beta is an OLS slope; report it with Newey–West standard errors
- ▣ Betas move over time: sector betas ranged from about -0.3 to 2.2 in rolling windows
- ▣ Extreme betas regress towards one: Blume and Vasicek shrinkage improve forecasts
- ▣ Romanian banks have the highest betas against BET; utilities and telecom the lowest
- ▣ Clean the data first: a single unadjusted corporate event destroys a regression

Jensen's Alpha and the GRS Test

- ▣ **Jensen's alpha** (Jensen (1968)): intercept of the market model, $\alpha_i = \mathbb{E}[R_i^e] - \beta_i \mathbb{E}[R_m^e]$; the CAPM implies $\alpha_i = 0$ for every asset
- ▣ One t -test per asset inflates false rejections (Section 7): **GRS**, Gibbons, Ross & Shanken (1989), tests $H_0 : \alpha_1 = \dots = \alpha_N = 0$ jointly

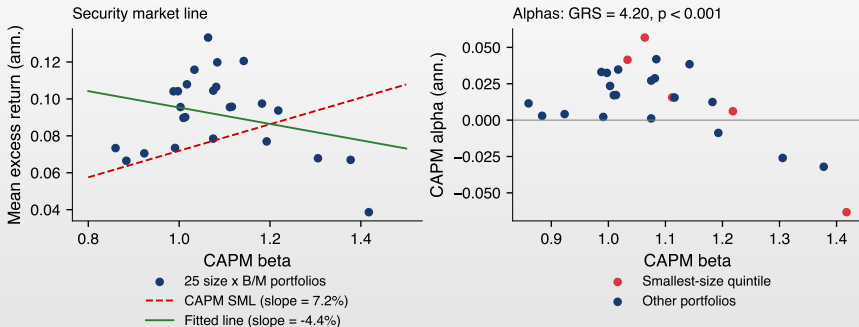
$$W = \frac{T - N - 1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$$

- ▣ $\hat{\Sigma}$: maximum-likelihood residual covariance (divided by T); $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market
- ▣ Exact, conditional on the factors, under i.i.d. multivariate Normal errors independent of the factors, with zero mean and constant nonsingular covariance; F : Fisher distribution
- ▣ Large W : the market is far from the mean–variance frontier of the test assets

The 25 Size and Book-to-Market Portfolios

- ▣ US stocks sorted each June into 5 **size** quintiles (market value) and 5 **B/M** quintiles
- ▣ $5 \times 5 = 25$ value-weighted portfolios: from small-growth (small, low B/M) to big-value
- ▣ Standard test assets since [Fama & French \(1993\)](#): they spread average returns widely
- ▣ Monthly, July 1963 – July 2026, $T = 757$
- ▣ Portfolios instead of single stocks: betas are estimated much more precisely

The Security Market Line in the Data



□ Left: average excess returns vs CAPM betas; right: alphas, smallest-size quintile in red

Reading the Security Market Line

- ▣ The CAPM predicts a line with slope equal to the premium, 7.2% a year
- ▣ The fitted cross-sectional line has slope -4.4% : higher beta, **lower** average return
- ▣ Betas span only 0.86–1.42, while average excess returns span 3.9%–13.3%
- ▣ Small-value alpha $+5.7\%$, small-growth alpha -6.3% a year; 10 of 25 alphas have $|t| > 1.96$
- ▣ $GRS = 4.20$, p -value 8×10^{-11} : the CAPM is rejected

GRS as a Sharpe-Ratio Test

- Identity (Gibbons, Ross & Shanken (1989)): with $\hat{\Sigma}$ and the moments estimated by maximum likelihood

$$\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha} = \widehat{SR}^2(f, R) - \widehat{SR}^2(f)$$

- $\widehat{SR}^2(f, R)$: squared Sharpe ratio of the tangency portfolio of the factor and the N test assets

- So GRS measures the Sharpe-ratio gain from the test assets

$$W = \frac{T - N - 1}{N} \cdot \frac{\widehat{SR}^2(f, R) - \widehat{SR}^2(f)}{1 + \widehat{SR}^2(f)}$$

- 25 portfolios, monthly Sharpe ratios: market 0.134 (0.47 a year); market plus portfolios 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1 + 0.134^2) = 4.20$; under H_0 , $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$, so the expected gain is only $N(1 + \widehat{SR}^2(f))/(T - N - 3) = 0.035$, against 0.146 observed

When the GRS Assumptions Fail

- ▣ **Heavy tails and volatility clustering** (Chapter 1): the F distribution is no longer exact
 - ▶ GMM Wald test: moments $\varepsilon_t \otimes (1, f_t)$, $W_{\text{GMM}} = \hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \rightarrow \chi_N^2$, with $\hat{V}_\alpha$ the HAC estimate of $\text{Var}(\hat{\alpha})$: robust to heteroskedasticity and autocorrelation
 - ▶ Residual bootstrap under H_0 : resample whole rows of residuals, rebuild returns with $\alpha = 0$ (Seminar B11); robust to non-Normality only if errors are i.i.d. and independent of the factors
- ▣ **Large N** : $\hat{\Sigma}$ has rank at most $T - 2$, so GRS is not computable when $N \geq T - 1$
 - ▶ Example: 100 size $\times$ B/M portfolios on 8 years of monthly data ($T = 96$); power falls well before that point
- ▣ Tests for many assets that avoid $\hat{\Sigma}^{-1}$: [Pesaran & Yamagata \(2024\)](#); power enhancement by screening large alphas: [Fan, Liao & Yao \(2015\)](#)

A Flat SML: Black's Insight



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia (2009 upload); public domain; Wikimedia Commons

- Black (1972): if borrowing is restricted, investors who want more risk buy high-beta assets instead of using leverage
- High-beta assets become expensive, so their future returns are lower
- Prediction: the SML is **flatter** than the CAPM line and starts at a **zero-beta** rate above R_f
- The same Black co-authored the Black–Scholes option formula (Chapter 12)

Betting Against Beta

- Frazzini & Pedersen (2014) turn the flat SML into a trading strategy, **BAB** (Betting Against Beta)

$$r_t^{\text{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- L : low-beta portfolio, leveraged to beta 1; H : high-beta portfolio, de-leveraged to beta 1
- The portfolio has beta zero by construction, yet earns a positive average return in their data
- Explanation: leverage constraints of investors, as in Black (1972)

Roll's Critique

- ▣ Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth
- ▣ Stock indices omit bonds, real estate, private firms and human capital
- ▣ Mathematical fact: an efficient portfolio prices all assets exactly by beta: with R_f on the CML, with a zero-beta rate (Black (1972)) on the risky-asset frontier
- ▣ Hence a CAPM test is a joint test of the model and of the efficiency of the proxy
- ▣ Practical answer: evaluate models by how well they price assets, not as literal truths

Low Power: GRS on Sector ETFs

- ▣ Same test on the 9 sector ETFs, monthly 1999–2026 ($T = 331$)
- ▣ $GRS = 0.53$, $p = 0.85$: the CAPM is **not** rejected; the sectors raise the annual Sharpe ratio only from 0.51 to 0.66
- ▣ Alphas: utilities +3.4%, energy +2.4%, financials −2.3% a year, all statistically weak
- ▣ Sectors mix small and big, value and growth: they do not spread the characteristics that matter
- ▣ Lesson: failing to reject is weak evidence; choose test assets with dispersion in expected returns

Recap: Testing the CAPM

- ▣ The CAPM says all alphas are zero; the GRS test checks this jointly
- ▣ On the 25 size $\times$ B/M portfolios, 1963–2026, the CAPM is strongly rejected
- ▣ The empirical SML is flat or negative; leverage constraints (Black; BAB) offer one explanation
- ▣ Roll: every test also tests the market proxy
- ▣ Test assets matter: sector ETFs cannot reject the CAPM

Cross-Sectional Tests: Fama and MacBeth



Eugene Fama receiving the Morgan Stanley–AFA
Award, 2008

Photo: [TonyTheTiger \(2008\)](#); CC BY-SA 3.0;
[Wikimedia Commons](#)

- **Fama & MacBeth (1973)**: estimate the **price of risk** λ of each factor
- **Risk premium** λ_k : extra expected return per unit of exposure $\beta_{i,k}$
- Model: $\mathbb{E}[R_i^e] = \lambda_0 + \beta_i^\top \lambda$
- Allows factors that are **not** traded portfolios, and firm characteristics as regressors
- Fama: Nobel Prize in Economics 2013

The Two Passes

- ▣ **Pass 1** (time series): for each asset i , estimate the loadings

$$R_{i,t}^e = a_i + \beta_i^\top \mathbf{f}_t + \varepsilon_{i,t} \quad \Rightarrow \quad \hat{\beta}_i$$

- ▣ **Pass 2** (cross section): for each month t , regress returns on the estimated betas

$$R_{i,t}^e = \lambda_{0,t} + \hat{\beta}_i^\top \boldsymbol{\lambda}_t + u_{i,t}, \quad i = 1, \dots, N$$

- ▣ Estimate: $\hat{\boldsymbol{\lambda}} = \frac{1}{T} \sum_t \hat{\boldsymbol{\lambda}}_t$, the average of T monthly slopes
- ▣ λ_0 : pricing error common to all assets; zero if the model is right

Fama–MacBeth Standard Errors

- Treat the monthly estimates $\hat{\lambda}_{k,t}$ as a sample of size T

$$se_{FM}(\hat{\lambda}_k) = \frac{s(\hat{\lambda}_{k,t})}{\sqrt{T}}, \quad s^2 = \frac{1}{T-1} \sum_t (\hat{\lambda}_{k,t} - \hat{\lambda}_k)^2$$

- Advantage: automatically robust to cross-sectional correlation of residuals
- If $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series instead
- Problem: the betas of Pass 2 are **estimated**, not known

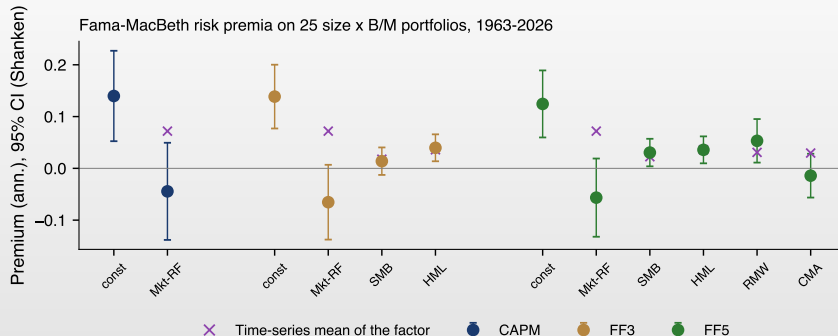
Errors in Variables: the Shanken Correction

- ▣ **Errors in variables:** using $\hat{\beta}$ instead of β makes se_{FM} too small
- ▣ The Fama–MacBeth variance already contains the factor term Σ_f/T ; [Shanken \(1992\)](#) scales only the rest by $(1 + c)$

$$\text{Var}_{Sh}(\hat{\lambda}) = (1 + c) \left[\text{Var}_{FM}(\hat{\lambda}) - \frac{1}{T} \Sigma_f \right] + \frac{1}{T} \Sigma_f, \quad c = \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda}$$

- ▣ Σ_f : covariance matrix of the factors; c : squared Sharpe ratio of the premia per period
- ▣ Small for FF3 here ($c = 0.03$ per month); potentially large when the estimated premia are large relative to the factor covariance, as for some non-traded factors; rescaling a factor leaves c unchanged
- ▣ Same result from **GMM** (Generalised Method of Moments, [Hansen \(1982\)](#)); see Quantinar

Estimated Risk Premia



25 size × B/M portfolios, 1963–2026; bars: 95% intervals with Shanken errors; crosses: factor means

Reading the Risk Premia

- ▣ CAPM: intercept 14.0% (se 4.4%), market premium -4.4% (se 4.8%)
- ▣ FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%); the market premium stays negative
- ▣ FF5: RMW 5.3% (2.1%); CMA -1.4% (2.2%) is not statistically distinguishable from zero (95% interval about $[-5.7\%, 2.9\%]$)
- ▣ For a traded factor, λ should equal the time-series factor mean (the crosses)
 - ▶ HML: 4.0% vs mean 3.6%, close; market: -6.5% vs mean 7.2%, far
- ▣ Market betas span only 0.86–1.42 (cross-sectional s.d. 0.14): λ_m is **weakly identified**, and a large intercept absorbs the level (Lewellen, Nagel & Shanken (2010))

Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model: all pricing errors $e = \mu - X\gamma$ are zero ($X = [1, B]$)
- GRS rejects here, so γ is a pseudo-true value; the noise in $\hat{B}$ multiplies e and adds a variance term ([Kan, Robotti & Shanken \(2013\)](#))
- Write both passes as one exactly identified GMM system; the sandwich variance with the full Jacobian is the robust one
 - ▶ Setting $e = 0$ in the Jacobian gives the correct-model, heteroskedasticity-robust version of [Jagannathan & Wang \(1998\)](#)
- FF5 on the 25 portfolios: s.e. of RMW 2.1% $\rightarrow$ 2.6%, CMA 2.2% $\rightarrow$ 2.7%, intercept 3.3% $\rightarrow$ 3.8%
- RMW: t falls from 2.5 to 2.1; HML changes by less than 0.05 percentage points

Worked Example: Four Standard Errors for the Value Premium

FF3, HML	$\hat{\lambda}$	s.e.	t	Significant at 5%?
Fama–MacBeth	3.97%	1.33%	2.99	yes
Newey–West (6 lags)	3.97%	1.62%	2.45	yes
Shanken	3.97%	1.33%	2.99	yes
Kan–Robotti–Shanken	3.97%	1.32%	3.00	yes

- Shanken barely moves the error: $c = 0.03$ per month
- Autocorrelation of the monthly slopes $\hat{\lambda}_t$ matters most here: Newey–West, 6 lags by the rule for $T = 757$
- At the $t > 3$ hurdle of Section 7, HML is borderline under every choice (2.45–3.00)

Useless and Weak Factors

- A **useless factor** g_t is independent of all returns: $\beta_g = 0$, so $\hat{\beta}_{i,g} = O_p(T^{-1/2})$ is pure noise
- Pass 2 divides by this noise: $\hat{\lambda}_g$ does not converge to zero and its t -test over-rejects when the model is misspecified ([Kan & Zhang \(1999\)](#))
- Simulation on our 25 portfolios, CAPM plus an independent Normal factor, 2000 replications, nominal level 5%
 - ▶ $\lambda_g = 0$ rejected in 60% (Fama–MacBeth errors) and 27% (Shanken errors) of the replications
 - ▶ First-pass Wald test of $\beta_{i,g} = 0$ for all i : rejected in 6.7%; run it before pass 2
- **Weak factors** (little dispersion in betas, like the market here): identification-robust tests ([Kleibergen \(2009\)](#)); with many assets, consistency of pass 2 depends on factor strength ([Anatolyev & Mikusheva \(2022\)](#))

A Skeptical Appraisal: Cross-Sectional R^2

- The 25 size $\times$ B/M portfolios have a strong three-factor structure
 - ▶ Any factors correlated with SMB and HML fit their average returns: a high OLS R^2 is a low hurdle (Lewellen, Nagel & Shanken (2010))
- Prescriptions of Lewellen, Nagel & Shanken (2010)
 - ▶ Add test assets with other characteristics, for example 30 industry portfolios
 - ▶ For traded factors impose $\lambda = \mathbb{E}[f]$ (the factor prices itself)
 - ▶ Report the **GLS** (Generalised Least Squares) R^2 , weighted by $\hat{V}^{-1}$ like GRS, and confidence intervals for R^2
- Individual stocks instead of portfolios: more dispersion in betas, but larger errors in variables (Ang, Liu & Schwarz (2020))

Worked Example: Adding 30 Industry Portfolios

Model	OLS R^2		GLS R^2	
	25	25 + 30	25	25 + 30
CAPM	0.08	0.00	0.06	0.01
FF3	0.66	0.22	0.21	0.08
FF5	0.76	0.24	0.26	0.10

- Monthly 1963–2026, free intercept and premia; $\hat{V}$: sample covariance of the test-asset returns
- FF3 falls from 66% to 22% once industries are added; the GLS R^2 never exceeds 26%
- With $\lambda = \mathbb{E}[f]$ and no intercept, the FF3 R^2 on the 55 assets is negative (-0.27)

Recap: Fama–MacBeth

- ▣ Pass 1 estimates betas; Pass 2 averages monthly cross-sectional slopes
- ▣ Fama–MacBeth errors handle cross-sectional correlation; Newey–West handles autocorrelation
- ▣ Shanken corrects for estimated betas (small for monthly traded factors); Kan–Robotti–Shanken also for misspecification
- ▣ Useless factors look priced in pass 2: test the first-pass betas first
- ▣ On the 25 portfolios the value and profitability premia are significant, but the cross-sectional R^2 collapses once industries are added

Arbitrage Pricing Theory

- ▣ **APT**: Arbitrage Pricing Theory, [Ross \(1976\)](#)
- ▣ Returns follow a K -factor model with small idiosyncratic risks

$$R_{i,t} = \mathbb{E}[R_i] + \beta_i^\top f_t + \varepsilon_{i,t}, \quad \mathbb{E}[f_t] = 0$$

- ▣ **No arbitrage** (no riskless profit without investment) implies, approximately

$$\mathbb{E}[R_i] - R_f = \beta_i^\top \lambda$$

- ▣ The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios

- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- Zero net investment: the return is a pure spread, $f_t = R_{\text{high},t} - R_{\text{low},t}$
- Fama–French 2×3 sort: size (small/big) × characteristic (30% / 40% / 30%)
 - ▶ **SMB** (Small Minus Big): small stocks minus big stocks
 - ▶ **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- Portfolios rebalanced each June using accounting data of the previous year

The Fama–French Models

- Fama & French (1992): size and B/M explain the cross section; beta does not
- **FF3** (three-factor model), Fama & French (1993)

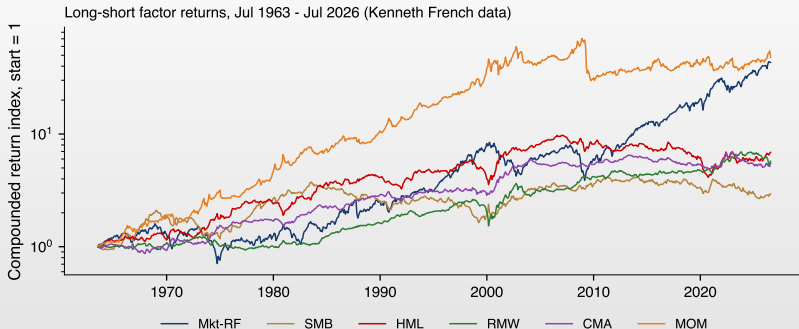
$$R_{i,t}^e = \alpha_i + b_i \text{MKT}_t + s_i \text{SMB}_t + h_i \text{HML}_t + \varepsilon_{i,t}$$

- **FF5** (five-factor model), Fama & French (2015), adds
 - ▶ **RMW** (Robust Minus Weak): high minus low operating profitability
 - ▶ **CMA** (Conservative Minus Aggressive): low minus high asset growth (investment)
- FF3 uses its published SMB (2×3 sorts on size and B/M); the FF5 SMB averages size spreads from B/M, profitability and investment sorts, so FF3 is not nested in FF5
- Test of a model: time-series alphas close to zero, GRS not rejected

Momentum, Carhart and the q-Factor Model

- ▣ **Momentum**: past winners keep winning over 3–12 months, [Jegadeesh & Titman \(1993\)](#)
- ▣ **MOM** (UMD, Up Minus Down): stocks with high minus low return from month $t - 12$ to $t - 2$
- ▣ [Carhart \(1997\)](#) four-factor model: FF3 + MOM, standard for fund evaluation
- ▣ **q-factor model**, [Hou, Xue & Zhang \(2015\)](#): market, size, investment and profitability (ROE)
 - ▶ ROE: Return On Equity; motivated by firms' investment decisions
- ▣ FF5 and the q-model are close relatives; FF5 omits momentum

Six Factors over Sixty-Three Years

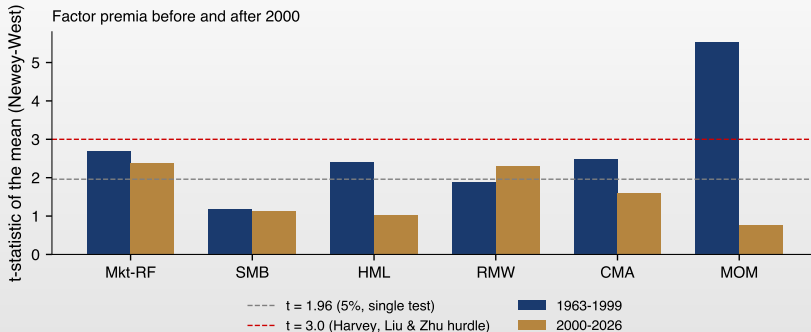


- Compounded monthly factor returns, starting at 1: excess or long-short returns, so no interest on the capital or collateral; log scale

Reading the Factor Growth

- ▣ July 1963 – July 2026: the compounded index reached 43.2 for Mkt-RF and 48.0 for MOM
- ▣ Value (HML) 6.9, profitability (RMW) 5.7, investment (CMA) 5.5, size (SMB) 2.9
- ▣ Momentum peaked before 2009 and lost heavily in the 2009 rebound: a **momentum crash**
- ▣ Value was flat or losing from about 2007 to 2020
- ▣ Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

Do the Premia Survive?



□ t -statistic of the mean monthly factor return (Newey-West), 1963–1999 vs 2000–2026

Reading the Split-Sample t -Statistics

- Momentum: $t = 5.53$ before 2000, 0.75 after; mean 10.7% vs 2.5% a year
- Value (HML): $t = 2.39$ then 1.02; mean 4.2% vs 2.8%
- Profitability (RMW) is the exception: $t = 1.90$ then 2.31
- Market premium robust in both halves ($t = 2.70$ and 2.39)
- McLean & Pontiff (2016): anomaly returns are 58% lower after academic publication, consistent with this decay

Factors as a Stochastic Discount Factor



Lars Peter Hansen, Nobel Prize in
Economics 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0;
[Wikimedia Commons](#)

- **SDF** (Stochastic Discount Factor) m_t : prices all excess returns, $\mathbb{E}[m_t R_{i,t}^e] = 0$
- A linear factor model is an SDF $m_t = 1 - \mathbf{b}^\top (\mathbf{f}_t - \mathbb{E}\mathbf{f})$
- These moment conditions are estimated by GMM, [Hansen \(1982\)](#)
- Survey of the discount-rate view: [Cochrane \(2011\)](#)
- Hansen shared the 2013 Nobel Prize with Fama and Shiller

Comparing Models: Maximum Sharpe Ratios

Max. Sharpe ratio (annual)	CAPM	FF3	Carhart	FF5	FF5 + MOM
1963–2026 ($T = 757$)	0.47	0.66	0.97	0.97	1.16
2000–2026 ($T = 319$)	0.48	0.54	0.67	0.97	1.02

- Barillas & Shanken (2018): judge a model by $SR^2(f) = \mu_f^\top \Sigma_f^{-1} \mu_f$ of its factors; the test assets do not matter for the comparison
- Nested models: in-sample SR^2 can only rise, so test the alpha of the added factor on the others (spanning regression)
 - HML on the other five factors: $\alpha = 0.9\%$ a year ($t = 0.75$); since 2000 -1.7% ($t = -0.81$): no significant incremental contribution of HML, as in Fama & French (2015)
- Non-nested FF5 vs Carhart: difference in annual SR^2 0.00, 95% interval $[-0.57, 0.66]$; since 2000 0.50, $[-0.24, 1.48]$ (stationary bootstrap, Politis & Romano (1994))

Beyond Sharpe Ratios: HJ Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997)): least-squares distance of the model SDF from the set of valid SDFs

$$\delta^2 = \min_b e(b)^\top \mathbb{E}[R^e R^{e\top}]^{-1} e(b), \quad e(b) = \mathbb{E}[m_t(b) R_t^e]$$

- ▶ Excess returns R_t^e with the normalisation $\mathbb{E}[m] = 1$ of the previous slide; for gross returns R_t : $e = \mathbb{E}[mR] - 1$, and the SDF needs a free scale ($\mathbb{E}[m] = 1/(1 + R_f)$)
- ▶ Same weighting matrix for every model, so distances are comparable; the test statistic has a weighted χ^2 limit
- Zoo-scale selection: Bayesian averaging over two quadrillion factor models, robust to weak factors (Bryzgalova, Huang & Julliard (2023))
- SDF shrinkage on characteristic-based factors: a few principal components of anomaly returns capture the SDF (Kozak, Nagel & Santosh (2020))
- Common lesson: compare models by a single, model-independent metric, with its sampling uncertainty

Recap: Multifactor Models

- The APT justifies several priced factors but does not name them
- Factors are long–short portfolios sorted on characteristics: SMB, HML, RMW, CMA, MOM
- FF3, Carhart, FF5 and the q-model are the standard benchmarks
- Momentum and value were much weaker after 2000; profitability stayed strong
- Factor models are SDFs; compare them by the Sharpe ratio of their factors (HML adds no significant alpha to the other FF5 factors and MOM) or by the HJ distance

The Factor Zoo

- ▣ Cochrane (2011) called the growing list of new factors a “zoo”
- ▣ Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- ▣ Each was tested as if it were the only one: $|t| > 1.96$ at 5%
- ▣ With many tests, some pass by chance: **data mining**, also called p-hacking
- ▣ Their proposal: a new factor needs $|t| > 3.0$

False Discovery Rate: Benjamini–Hochberg

- ▣ **FDR** (False Discovery Rate): expected share of false rejections among all rejections
- ▣ **BH** procedure (Benjamini & Hochberg (1995)): find the largest j with

$$p_{(j)} \leq \frac{j}{M} \alpha$$

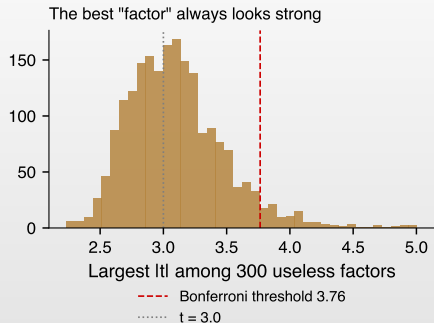
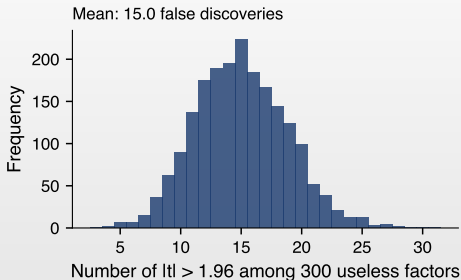
and reject $H_{(1)}, \dots, H_{(j)}$

- ▣ Controls FDR at α for independent tests, or under positive regression dependence on the true nulls (Benjamini & Yekutieli (2001))
 - Under arbitrary dependence: the Benjamini–Yekutieli variant divides the thresholds by $\sum_{j=1}^M 1/j$
- ▣ Less strict than **FWER** (Family-Wise Error Rate) control by Bonferroni or Holm (Holm (1979)), defined in Chapter 2: suited to searching many candidate factors

Worked Example: Ten Candidate Factors

- Sorted p -values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62; $\alpha = 5\%$
- Naive ($p \leq 0.05$): **7** discoveries
- Bonferroni ($p \leq 0.005$): **2**
- Holm: $0.0004 \leq 0.005$, $0.002 \leq 0.0056$, $0.006 \leq 0.00625$, $0.011 > 0.0071$ stop: **3**
- BH thresholds $0.005j$: $p_{(6)} = 0.027 \leq 0.030$ is the last success: **6**

Simulation: 300 Useless Factors



- 2000 replications; each draws 300 independent t -statistics from the standard Normal distribution (no true premium)

Reading the Simulation

- ▣ On average 15.0 useless factors pass $|t| > 1.96$; expected value $300 \times 0.05 = 15$
- ▣ On average 0.83 pass $|t| > 3$; at least one passes in 57% of the replications
- ▣ The median largest $|t|$ is 3.06: the best “factor” always looks strong
- ▣ Bonferroni threshold for 300 tests: $|t| > 3.76$
- ▣ Hurdles rise with the number of tests actually tried, including unpublished ones

Taming the Zoo and the Replication Debate

- [Feng, Giglio & Xiu \(2020\)](#): does a new factor add information given the existing zoo?
 - ▶ **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator): penalised regressions choose the controls among hundreds of factors
 - ▶ Most new factors are redundant; only a few add pricing information
- [Jensen, Kelly & Pedersen \(2023\)](#): a Bayesian view of 153 factors in 93 countries
 - ▶ Most factors replicate; they cluster into 13 themes; out-of-sample and international evidence is consistent
- The two views agree: the zoo is highly redundant, and a few themes carry the information

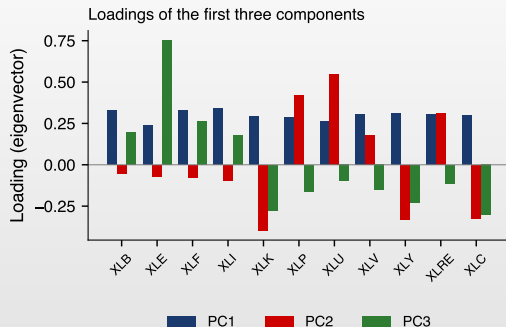
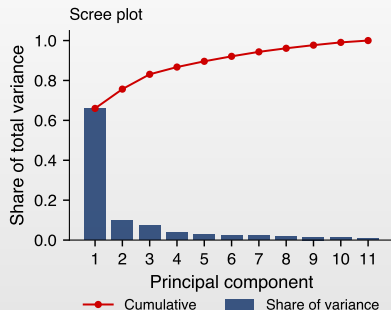
Recap: The Factor Zoo

- ▣ Hundreds of published factors imply many false discoveries at $|t| > 1.96$
- ▣ Bonferroni and Holm control the chance of any false discovery; BH controls their share
- ▣ With 300 useless factors, the best one has $|t| \approx 3$: raise the hurdle
- ▣ Redundancy is the main problem: few themes carry most of the information

Prerequisites Recap: Principal Component Analysis

- ▣ **Latent factors**: estimated from returns alone, without naming them in advance
- ▣ **PCA** (Principal Component Analysis) of the correlation matrix of standardised returns: $C = \Gamma \Lambda \Gamma^\top$, components $y_t = \Gamma^\top z_t$
- ▣ Eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$; eigenvectors are the **loadings**; component k explains λ_k/N of the variance
- ▣ Two assets with correlation ρ : $\lambda = 1 \pm \rho$, eigenvectors $(1, 1)/\sqrt{2}$ (the “market”) and $(1, -1)/\sqrt{2}$ (the spread); in a crisis ρ rises and PC1 dominates
- ▣ Textbook: [Härdle & Simar \(2019\)](#), chapter on principal components; more examples in Seminar A8

PCA of the US Sector ETFs



11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ($T = 2073$ common days)

Reading the Principal Components

- ▣ PC1 explains 66.0% of the variance; all loadings are positive (0.24–0.34)
 - ▶ Correlation of PC1 with SPY: 0.954; the first latent factor is the market
- ▣ PC2 (9.7%): utilities +0.55, staples +0.42, real estate +0.31 vs technology –0.40, consumer discretionary –0.33
 - ▶ A **defensive-versus-growth** factor, with no name given in advance
- ▣ PC3 (7.4%) is dominated by energy; three components explain 83.1%
- ▣ Only 2 eigenvalues exceed 1 (Kaiser rule): a heuristic, not a consistent estimator of the number of factors (next slides)

How Many Factors? Consistent Estimators

- Approximate factor model $x_{it} = \lambda_i^\top F_t + e_{it}$ with $N, T \rightarrow \infty$; $V(k)$: mean squared residual after the first k principal components
- Bai & Ng (2002): $\hat{k} = \arg \min_k \left[\ln V(k) + k \frac{N+T}{NT} \ln \min(N, T) \right]$ (criterion IC_{p2}), consistent as $N, T \rightarrow \infty$
- Ahn & Horenstein (2013): eigenvalue ratio $\hat{k} = \arg \max_k \mu_k / \mu_{k+1}$, no penalty to choose
- Onatski (2010): separates factor eigenvalues from the edge of the noise eigenvalue distribution
- Kaiser ($\lambda_k > 1$) has no asymptotic justification: the noise eigenvalues of a correlation matrix scatter around 1

Worked Example: The Number of Factors in Our Data

Panel	N	T	Kaiser	Bai–Ng	Ahn–Horenstein
11 sector ETFs, daily 2018–2026	11	2073	2	5	1
100 size $\times$ B/M portfolios, monthly 1963–2026	95	757	3	4	1

- Standardised returns; maximum k : 5 for sectors, 8 for portfolios; 5 portfolios with missing months are dropped
- Ahn–Horenstein picks the dominant market factor (ratio 14.6); its next peak is at $k = 3$ (ratio 3.5), the FF3 structure
- Bai–Ng on 11 sectors hits the upper bound $k = 5$: the penalty is built for large N , so with small N the answer is fragile

Limits of PCA for Asset Pricing

- ▣ Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- ▣ Components explain **variance**, not necessarily **average returns**
- ▣ Individual stocks enter and exit the sample: a stock panel is unbalanced
- ▣ Component signs are arbitrary; rotations preserve principal components only within repeated-eigenvalue eigenspaces, while a latent-factor model is identified only up to rotation
- ▣ PCA and factor analysis in finance: [Franke, Härdle & Hafner \(2019\)](#); multivariate background: [Härdle & Simar \(2019\)](#)

Risk-Premium PCA and Three-Pass Premia

- ▣ **RP-PCA** (Risk-Premium PCA, [Lettau & Pelger \(2020\)](#)): principal components of $\frac{1}{T}X^T X + \gamma \bar{x}\bar{x}^T$
 - ▶ $\gamma = -1$ gives covariance PCA; $\gamma > -1$ also rewards factors that explain average returns
 - ▶ In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- ▣ **Three-pass estimator** ([Giglio & Xiu \(2021\)](#)): the premium of an observed factor g_t when the model omits factors
 - ▶ (1) PCA of test-asset returns; (2) cross-sectional regression of means on the latent betas; (3) regress g_t on the latent factors and map their premia to g_t
- ▣ Both keep PCA's statistical backbone but target pricing, not only variance; IPCA (next) adds characteristics

Instrumented PCA

- ▣ **IPCA:** Instrumented Principal Component Analysis, [Kelly, Pruitt & Su \(2019\)](#)
- ▣ Loadings are functions of L observed characteristics $\mathbf{z}_{i,t}$ (size, B/M, momentum, ...)

$$R_{i,t+1}^e = \alpha_{i,t} + \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_{t+1} + \varepsilon_{i,t+1}, \quad \boldsymbol{\beta}_{i,t}^\top = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\beta$$

- ▣ $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the latent factors $\mathbf{f}_t$ are estimated jointly by alternating least squares
- ▣ Test: are characteristics priced through betas ("covariances"), or directly as alphas ($\alpha_{i,t} = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\alpha$)?

What IPCA Teaches

- Title of the paper: “Characteristics are covariances”
- In their US stock data, a few IPCA factors price the cross section better than observable-factor models with as many factors
- The alpha term Γ_α becomes insignificant: characteristics predict returns because they measure risk exposure
- Links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors price the cross section of coins (Liu, Tsyvinski & Wu (2022)); dimension reduction on crypto and traditional return features: Pele et al. (2023), Jheng et al. (2026)

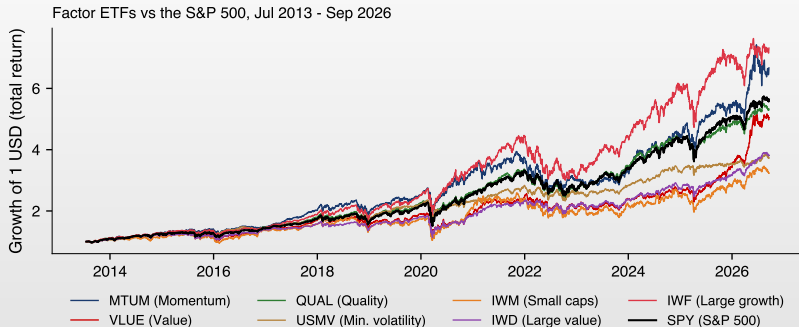
Recap: Latent Factors

- PCA finds factors as eigenvectors of the correlation matrix; choose their number with Bai–Ng or Ahn–Horenstein, not Kaiser
- In sector ETFs, PC1 is the market (66%), PC2 a defensive-versus-growth factor (10%)
- PCA explains variance, with fixed loadings; it is not an asset pricing model by itself
- IPCA lets loadings depend on characteristics and tests whether characteristics are risk

Factor ETFs

- ▣ **Smart beta:** rule-based portfolios that tilt towards factors, sold as ETFs
- ▣ MTUM (momentum), VLU (value), QUAL (quality: high profitability, low leverage)
- ▣ USMV (minimum volatility: the low-risk anomaly), IWM (small caps, Russell 2000)
- ▣ IWD / IWF: Russell 1000 value / growth (large caps)
- ▣ Long-only: they keep market exposure and tilt towards selected factors; they need not replicate the long leg of an academic factor or have a market beta close to 1

Factor ETFs versus the Market

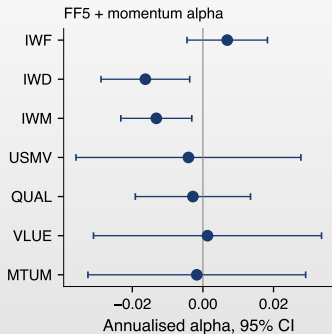


□ Growth of 1 USD with dividends re-invested, from 18 July 2013 (first common day)

Reading the Factor ETF Returns

- ▣ **CAGR:** large growth (IWF) 16.3% and momentum (MTUM) 15.5% beat SPY (14.0%)
- ▣ Quality 13.5%, value 13.1%, large value 10.7%, minimum volatility 10.5%, small caps 9.3%
- ▣ This period favoured large growth stocks: value and small caps lagged
- ▣ Thirteen years of one realisation: not evidence against the premia on its own

Alphas of Factor ETFs



Factor loadings

	Mkt-RF	SMB	HML	RMW	CMA	MOM
MTUM	1.03	-0.12	0.01	-0.22	-0.05	0.36
VLUE	0.98	0.08	0.34	-0.06	0.09	-0.03
QUAL	0.97	-0.08	-0.04	0.15	0.03	-0.04
USMV	0.74	-0.11	-0.02	0.20	0.21	-0.02
IWM	1.04	0.85	0.08	-0.10	-0.02	0.03
IWD	0.94	-0.04	0.29	0.09	0.19	-0.04
IWF	1.03	-0.11	-0.28	-0.03	-0.12	-0.00

□ Daily regressions on FF5 + MOM with Newey–West errors, each ETF on its own trading calendar (2000–2026)

 MFM_ch3_etf_alphas

Reading the ETF Alphas

- ▣ The loadings match the labels: MTUM on MOM 0.36, VLUU on HML 0.34, IWM on SMB 0.85
- ▣ Growth (IWF) loads -0.28 on HML; USMV has market beta 0.74 and positive RMW, CMA
- ▣ Factor-model alphas are close to zero: MTUM -0.2% ($t = -0.11$), QUAL -0.3%
- ▣ Russell ETFs: IWM -1.3% ($t = -2.57$), IWD -1.6% ($t = -2.55$): costs and index construction
- ▣ R^2 between 0.81 and 0.97: the ETFs are mostly repackaged factor exposure

Factor Timing and Crowding

- **Factor timing:** switching between factors based on signals (valuation spreads, recent returns)
 - ▶ Noisy premia and short histories make timing hard to verify out of sample (Chapter 8)
- **Crowding:** when many investors hold the same factor positions
 - ▶ Premia can shrink (McLean & Pontiff (2016)), and crowded positions can amplify losses during forced unwinding; the 2009 momentum crash coincided with a sharp rebound of past losers (Daniel & Moskowitz (2016))
- **Costs:** long-short factors ignore trading costs and short-selling fees; ETFs pay them

Recap: Factor Investing

- ▣ Factor ETFs deliver the promised exposures and keep most of the market exposure (betas 0.74–1.04)
- ▣ After controlling for the factors, their alphas are zero or slightly negative
- ▣ 2013–2026 rewarded growth and momentum, punished value and size
- ▣ Timing and crowding are open problems; evaluate them out of sample

Case Study: Research Question

- **The challenge** to factor research takes two forms (Jensen, Kelly & Pedersen (2023), Introduction)
 - ▶ No internal validity: results cannot be reproduced; Hou, Xue & Zhang (2020) replicate only 35% of their anomalies
 - ▶ No external validity: results come from testing many factors without controlling the FDR; Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Question:** do published factors have significant CAPM alphas in the original data, out of sample and in other countries, once multiple testing is taken into account?
- **Why it matters**
 - ▶ If most factors are false discoveries, multifactor models and factor investing (Sections 6 and 9) rest on noise
 - ▶ The factor zoo (Section 7) is treated as a statistical problem; the paper treats it as correlated evidence on a few economic themes
- **Link to this chapter:** Jensen's alpha (Section 4) is the object tested; Benjamini–Yekutieli (Section 7) is the frequentist benchmark; Bayesian shrinkage extends the Vasicek beta (Seminar 3, A3) to alphas

Case Study: Data and Method

- **Data (Section II):** 153 factors in 93 countries; 138 from the list of [Hou, Xue & Zhang \(2020\)](#) plus 15 from the wider literature; US returns from 1926, most developed markets from 1986; all in US dollars, in excess of the US Treasury bill
 - ▶ Each country and month: terciles of the characteristic, breakpoints from non-micro stocks; "capped" value weights (market equity winsorised at the NYSE 80th percentile); long the tercile with the higher expected return in the original paper, one-month holding
 - ▶ At least 5 stocks per leg and 60 monthly observations; every factor scaled to a 10% annual idiosyncratic volatility
 - ▶ 13 themes: hierarchical clustering of US factors, distance 1 – correlation of CAPM residuals, Ward linkage (Section II.B)
- **Model (Section I):** $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$; alphas $\alpha_i = c_j + s_n + w_i$ (theme j , signal n , factor i ; eq. 21), prior mean 0
 - ▶ Prior variance $\Omega = MM^\top \tau_c^2 + ZZ^\top \tau_s^2 + I \tau_w^2$ (eq. 23); posterior mean $(\Omega^{-1} + T\Sigma^{-1})^{-1} T\Sigma^{-1} \hat{\alpha}$ (Proposition 4)
 - ▶ One factor: $E(\alpha | \hat{\alpha}) = \kappa \hat{\alpha}$, $\kappa = 1/(1 + \sigma^2/(\tau^2 T))$ (eq. 5): short, noisy samples are shrunk more
 - ▶ Empirical Bayes: τ_c, τ_s, τ_w by maximum likelihood of $\hat{\alpha} \sim N(0, \Omega + \hat{\Sigma}/T)$ (Section I.C.2)
- **Replication rate:** share of factors significant in the original paper whose alpha is significant here; Bayesian discovery if posterior mean / posterior s.d. ≥ 1.96 (eq. 26)

Case Study: Key Results

Replication rates (Figure 1)

Hou, Xue & Zhang (2020)	35.0%
US, raw return	55.6%
originally significant	61.3%
CAPM alpha	82.4%
Benjamini–Yekutieli	75.6%
Empirical Bayes	82.4%
World, empirical Bayes	82.4%

- OLS: 98 of 119 US factors; Benjamini–Yekutieli needs $|t| > 2.7$ (Figure 4)
 - ▶ $\Rightarrow$ testing alphas, not raw returns, drives most of the gap with Hou, Xue & Zhang (2020)
- Bayesian FDR 0.1%, 95% interval $[0, 1\%]$; expected share of true factors 94% (Section III.C)
 - ▶ $\Rightarrow$ shrinkage controls false discoveries without the power loss of a higher t hurdle
- Mega and large stocks: 77.3% and 79.8% (Figure 5); 11 of 13 themes at least 50%, all but low leverage and size
 - ▶ $\Rightarrow$ not an artefact of micro-caps
- Mean alpha 0.49% a month in sample, 0.26% after the original sample: -47% ; out-of-sample on in-sample slope 0.26 after the sample (Figure 8), against about 0.9 expected from shrinkage alone
 - ▶ $\Rightarrow$ decay is expected, but the strongest factors decay more: arbitrage or data mining
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

Case Study: Replicate, Limits, Extend

- **Replicate with course data** (Seminar 3, C3): US long–short portfolios of the [Kenneth French Data Library](#) on 15 characteristics, monthly from July 1963; CAPM alphas on Mkt–RF, scaled to 10% idiosyncratic volatility; OLS, Benjamini–Yekutieli and empirical Bayes replication rates; Ward clusters
- Data and code of the paper: [jpkfactors.com](#), [github.com/bkelly-lab/ReplicationCrisis](#)
- **Limits stated by the authors**
 - ▶ CAPM is the only benchmark: low-leverage factors fail on CAPM alpha but matter jointly with other themes (Section III.A)
 - ▶ Characteristics are built consistently, sometimes unlike the original paper (Section II.C)
 - ▶ Unobserved factors bias the prior: with the scenarios of [Harvey, Liu & Zhu \(2016\)](#) the US rate falls to 81.5% and 79.8% (Figure 10)
- **Limits of the course version:** 15 US signals instead of 153 in 93 countries; library breakpoints instead of non-micro terciles with capped weights; one region, so $\tau_s = 0$
- **Open extension:** the paper's data end in 2020; did the posterior alphas and the replication rate hold in 2021–2026, and can a prior estimated on US factors sharpen short-sample BVB factors (Seminar 3, C1)?

An Open Question: Why Do Factor Premia Decay after Publication?

- ▣ McLean & Pontiff (2016): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication
 - ▶ the gap, 32%, is attributed to publication-informed trading; the out-of-sample drop bounds data mining
- ▣ Jensen, Kelly & Pedersen (2023): in a Bayesian model that shrinks in-sample premia, most of 153 factors replicate, in the US and in 93 countries
- ▣ Chen & Zimmermann (2022): open-source data and code reproducing the published cross-sectional predictors
- ▣ Open: how much of the decay is arbitrage, data mining or a change in risk, and does it differ across factors and markets?

The Discovery Loop, Step by Step (1/2)

▣ Literature

- ▶ Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.
- ▶ Resolve every DOI on Crossref; check each headline number against the abstract

▣ Hypothesis

- ▶ Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.

▣ Data and code

- ▶ Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.
- ▶ Replicate a known number first: momentum $t = 5.53$ before 2000 and 0.75 after (split-sample chart)

The Discovery Loop, Step by Step (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).

▣ Critique

- ▶ Referee this: one factor, one publication date, $t = -1.45$. What is the power to detect a 58% decay, and why does pooling many anomalies change it?

▣ Report

- ▶ In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
- ▶ Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
- ▶ Prompts that shaped a decision go into AI_USE.md

- ▣ The assistant enumerates specifications; the human fixes which ones count before running them

What the Researcher Must Check

- **Wrong event date:** sample end, working paper and journal issue are different dates
 - ▶ test: read them from the paper itself; Jegadeesh–Titman: sample ends 1989, published March 1993
- **Choosing the factor after seeing it decay:** fix the set of factors in advance; Holm or BH across them
- **Serially correlated, crash-prone returns:** Newey–West errors; rerun without 2009
- **Data vintages:** the factor library is revised retroactively
 - ▶ test: record the download date; reproduce the chapter's split-sample t -statistics
- **Misquoted literature:** a common slip gives 58% as the out-of-sample decline; it is post-publication (26% out of sample)
- **Low power:** a single factor rarely rejects; report the confidence interval of the decay, not only its sign

Mini-Case: Was Momentum Arbitraged Away?

- A typical assistant proposal: “momentum disappeared once [Jegadeesh & Titman \(1993\)](#) published it”
- Design of [McLean & Pontiff \(2016\)](#): $MOM_t = a + b_1 \text{PostSample}_t + b_2 \text{PostPub}_t + e_t$, monthly, 1965–2026, Newey–West errors
 - ▶ post-sample: 1990-01 – 1993-03; post-publication: from 1993-04; $H_0 : b_2 = 0$ against $H_1 : b_2 < 0$
 - ▶ b_2 : total decline relative to the in-sample mean; $b_2 - b_1$: incremental effect of publication
- In-sample mean $a = 0.82\%$ a month ($t = 4.32$)
- Total decline $b_2 = -0.45\%$ ($t = -1.45$; 95% interval $[-1.05\%, 0.16\%]$): mean 0.37% a month, 54% lower
- Incremental $b_2 - b_1 = -0.79\%$ (s.e. 0.68%, $t = -1.18$); placebo, Mkt-RF: $b_2 = +0.40\%$ ($t = 1.12$)
- Interpretation: the point estimate is close to the 58% of [McLean & Pontiff \(2016\)](#), but one factor cannot reject H_0 at 5%
 - ▶ arbitrage is one interpretation, not an identified cause; pooling 97 predictors raises precision, while sample, cross-predictor dependence and heterogeneity also differ

 ai_discovery_case.py

Your Turn: A Project Seed

- **Question:** do the Fama–French factors and momentum decay after publication, and does the decay differ across factors?
 - ▶ pooled regression of SMB, HML, RMW, CMA and MOM on factor-specific post-sample and post-publication dummies (Mkt-RF: placebo, no publication date); errors robust to cross-factor correlation and autocorrelation
 - ▶ whether decay is larger where arbitrage is easier needs stock-level proxies (size, liquidity, short-sale costs), not factor returns
- **Replicate first:** the chapter's split-sample t -statistics and the mini-case estimates
- **Deliverables**
 - ▶ a pre-registration note (H_0 , H_1 , dates from each paper, factor set, corrections) committed before the results
 - ▶ code as a Quantlet; a report with Newey–West inference, placebo periods and power
- **Logs:** prompts in `AI_USE.md`; every caught error (a wrong publication date, a misquoted number) in `AI_ERRORS.md`

Conclusions

- ▣ The CAPM is elegant but fails on the 25 size $\times$ B/M portfolios: the SML is flat (slope -4.4%), $GRS = 4.20$
- ▣ Betas must be estimated with robust errors, change over time and should be shrunk
- ▣ Two-pass premia need errors robust to estimated betas and to misspecification; high R^2 on 25 portfolios is a low hurdle
- ▣ Hundreds of factors require multiple-testing hurdles ($|t| > 3$); models are compared by the Sharpe ratio of their factors
- ▣ Latent factors: PCA recovers the market and a defensive-versus-growth factor; IPCA links them to characteristics
- ▣ Factor ETFs are cheap factor exposure, not alpha

Links to Other Chapters

- ▣ **Chapter 2** (market efficiency): an alpha is either an anomaly (inefficiency) or a missing risk factor; the **joint hypothesis** problem
- ▣ **Chapter 4** (portfolio optimisation): factor covariance models and shrinkage for Markowitz
- ▣ **Chapters 5–6**: time-varying volatility and correlation, conditional betas
- ▣ **Chapter 8**: backtesting factor strategies without overfitting
- ▣ **Chapter 13**: machine learning for the cross section of returns

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Generalized Method of Moments
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- ▣ Factor data: [Kenneth French Data Library](#)

References (1/7)

- ▣ Ahn, S.C., Horenstein, A.R. (2013). Eigenvalue ratio test for the number of factors. *Econometrica*, 81(3), 1203–1227.
- ▣ Anatolyev, S., Mikusheva, A. (2022). Factor models with many assets: strong factors, weak factors, and the two-pass procedure. *Journal of Econometrics*, 229(1), 103–126.
- ▣ Ang, A., Liu, J., Schwarz, K. (2020). Using stocks or portfolios in tests of factor models. *Journal of Financial and Quantitative Analysis*, 55(3), 709–750.
- ▣ Bai, J., Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221.
- ▣ Barillas, F., Shanken, J. (2018). Comparing asset pricing models. *Journal of Finance*, 73(2), 715–754.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Black, F. (1972). Capital market equilibrium with restricted borrowing. *Journal of Business*, 45(3), 444–455.
- ▣ Blume, M.E. (1971). On the assessment of risk. *Journal of Finance*, 26(1), 1–10.
- ▣ Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4), 1165–1188.
- ▣ Bryzgalova, S., Huang, J., Julliard, C. (2023). Bayesian solutions for the factor zoo: we just ran two quadrillion models. *Journal of Finance*, 78(1), 487–557.

References (2/7)

- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- ▣ Carhart, M.M. (1997). *On persistence in mutual fund performance*. *Journal of Finance*, 52(1), 57–82.
- ▣ Chen, A.Y., Zimmermann, T. (2022). *Open source cross-sectional asset pricing*. *Critical Finance Review*, 11(2), 207–264.
- ▣ Cochrane, J.H. (2011). *Presidential address: discount rates*. *Journal of Finance*, 66(4), 1047–1108.
- ▣ Daniel, K., Moskowitz, T.J. (2016). *Momentum crashes*. *Journal of Financial Economics*, 122(2), 221–247.
- ▣ Fama, E.F., French, K.R. (1992). *The cross-section of expected stock returns*. *Journal of Finance*, 47(2), 427–465.
- ▣ Fama, E.F., French, K.R. (1993). *Common risk factors in the returns on stocks and bonds*. *Journal of Financial Economics*, 33(1), 3–56.
- ▣ Fama, E.F., French, K.R. (2015). *A five-factor asset pricing model*. *Journal of Financial Economics*, 116(1), 1–22.
- ▣ Fama, E.F., MacBeth, J.D. (1973). *Risk, return, and equilibrium: empirical tests*. *Journal of Political Economy*, 81(3), 607–636.
- ▣ Fan, J., Liao, Y., Yao, J. (2015). *Power enhancement in high-dimensional cross-sectional tests*. *Econometrica*, 83(4), 1497–1541.

References (3/7)

- ▣ Feng, G., Giglio, S., Xiu, D. (2020). Taming the factor zoo: a test of new factors. *Journal of Finance*, 75(3), 1327–1370.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Frazzini, A., Pedersen, L.H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–25.
- ▣ Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- ▣ Giglio, S., Xiu, D. (2021). Asset pricing with omitted factors. *Journal of Political Economy*, 129(7), 1947–1990.
- ▣ Hansen, L.P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4), 1029–1054.
- ▣ Hansen, L.P., Jagannathan, R. (1997). Assessing specification errors in stochastic discount factor models. *Journal of Finance*, 52(2), 557–590.
- ▣ Härdle, W.K., Simar, L. (2019). *Applied Multivariate Statistical Analysis*, 5th ed. Springer.
- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

References (4/7)

- ▣ Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- ▣ Hou, K., Xue, C., Zhang, L. (2015). Digesting anomalies: an investment approach. *Review of Financial Studies*, 28(3), 650–705.
- ▣ Hou, K., Xue, C., Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- ▣ Jagannathan, R., Wang, Z. (1998). An asymptotic theory for estimating beta-pricing models using cross-sectional regression. *Journal of Finance*, 53(4), 1285–1309.
- ▣ Jegadeesh, N., Titman, S. (1993). Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- ▣ Jensen, M.C. (1968). The performance of mutual funds in the period 1945–1964. *Journal of Finance*, 23(2), 389–416.
- ▣ Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance?. *Journal of Finance*, 78(5), 2465–2518.
- ▣ Jheng, S.-L., Găman, Ș., Conda, A., Pele, D.T., Härdle, W.K., Teng, H.-W., Osterrieder, J. (2026). Crypto currency returns. SSRN Working Paper.
- ▣ Kan, R., Robotti, C., Shanken, J. (2013). Pricing model performance and the two-pass cross-sectional regression methodology. *Journal of Finance*, 68(6), 2617–2649.
- ▣ Kan, R., Zhang, C. (1999). Two-pass tests of asset pricing models with useless factors. *Journal of Finance*, 54(1), 203–235.

References (5/7)

- ▣ Kelly, B.T., Pruitt, S., Su, Y. (2019). Characteristics are covariances: a unified model of risk and return. *Journal of Financial Economics*, 134(3), 501–524.
- ▣ Kleibergen, F. (2009). Tests of risk premia in linear factor models. *Journal of Econometrics*, 149(2), 149–173.
- ▣ Kozak, S., Nagel, S., Santosh, S. (2020). Shrinking the cross-section. *Journal of Financial Economics*, 135(2), 271–292.
- ▣ Lettau, M., Pelger, M. (2020). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies*, 33(5), 2274–2325.
- ▣ Lewellen, J., Nagel, S., Shanken, J. (2010). A skeptical appraisal of asset pricing tests. *Journal of Financial Economics*, 96(2), 175–194.
- ▣ Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1), 13–37.
- ▣ Liu, Y., Tsyvinski, A., Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2), 1133–1177.
- ▣ Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- ▣ McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability?. *Journal of Finance*, 71(1), 5–32.

References (6/7)

- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Onatski, A. (2010). Determining the number of factors from empirical distribution of eigenvalues. *Review of Economics and Statistics*, 92(4), 1004–1016.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Pesaran, M.H., Yamagata, T. (2024). Testing for alpha in linear factor pricing models with a large number of securities. *Journal of Financial Econometrics*, 22(2), 407–460.
- ▣ Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Roll, R. (1977). A critique of the asset pricing theory's tests. Part I. *Journal of Financial Economics*, 4(2), 129–176.
- ▣ Ross, S.A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3), 341–360.
- ▣ Shanken, J. (1992). On the estimation of beta-pricing models. *Review of Financial Studies*, 5(1), 1–33.
- ▣ Sharpe, W.F. (1964). Capital asset prices: a theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.

References (7/7)

- ▣ Vasicek, O.A. (1973). A note on using cross-sectional information in Bayesian estimation of security betas. *Journal of Finance*, 28(5), 1233–1239.
- ▣ Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.